Haleigh Renfroe

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Education

Morehead State University – Bachelor of Science in Computer Science | *GPA: 3.3* May 2026

Skills

Languages: C, C++, Python, JavaScript, SQL, HTML, CSS
Frameworks/Libraries: React, Svelte, Pandas, Matplotlib, OpenCV, MediaPipe
Embedded Systems: STM32, Embedded C, Hardware Debugging, Soldering
Tools/Platforms: Git, Linux, SQLite, Postgres, Docker

Experience

Morehead State University Space Science Center, Morehead, KY
Space Systems Software Engineer June 2026 - Present

- Contributed to development of the Mission Operations Center GUI for NASA's Lunar Environment Monitoring Station (LEMS), supporting payload commanding, telemetry monitoring, and mission operations.
- Developed a Python CLI for the Science Operations Center to identify and prioritize requested LEMS seismology data files, reducing manual review of large datasets.
- Debugged and updated embedded C firmware and Python tooling for an STM32-controlled anechoic chamber antenna turntable, restoring reliable motion and test functionality.
- Developing antenna test automation software to control turntable position, capture peak frequency and magnitude measurements, and generate plots for analysis.

Morehead State University Space Science Center, Morehead, KY
Student Worker September 2025 - May 2026

- Developed Python tools to parse, validate, and convert raw lunar telemetry into structured, readable data.
- Built telemetry visualization interfaces using HTML, CSS, Jinja2, and the AMMOS Instrument Toolkit.
- Collaborated with research teams to translate scientific data requirements into software workflows.

Ashland Police Department – Ashland, KY
Digital Forensics/Information Technology Intern May 2025 - August 2025

- Performed digital evidence acquisition and forensic imaging while verifying data integrity through cryptographic hashes.
- Supported hardware deployments, system upgrades, and technical troubleshooting across the department.
- Assisted staff during critical IT incidents, including ransomware response.

Projects

Junior College Basketball Recruiting Database January 2026 - May 2026

- Contributed to development of a full-stack recruiting platform aggregating NJCAA player data into a centralized, searchable database for university athletics
- Built frontend features using React and Vite, enabling filtering and comparison of athlete data
- Implemented user-facing functionality including informational pages and account management.

Facial Landmark Authentication System August 2025 - May 2026

- Developed a real-time facial recognition authentication system using Python, OpenCV, and MediaPipe
- Implemented a geometric landmark–based recognition pipeline with face alignment, normalization, and sigmoid-weighted similarity scoring for robust identification
- Engineered security features including blink-based liveness detection, multi-face detection, and configurable security levels to reduce spoofing and unauthorized access
- Awarded **Most Outstanding Senior Research Project** for technical innovation and system design